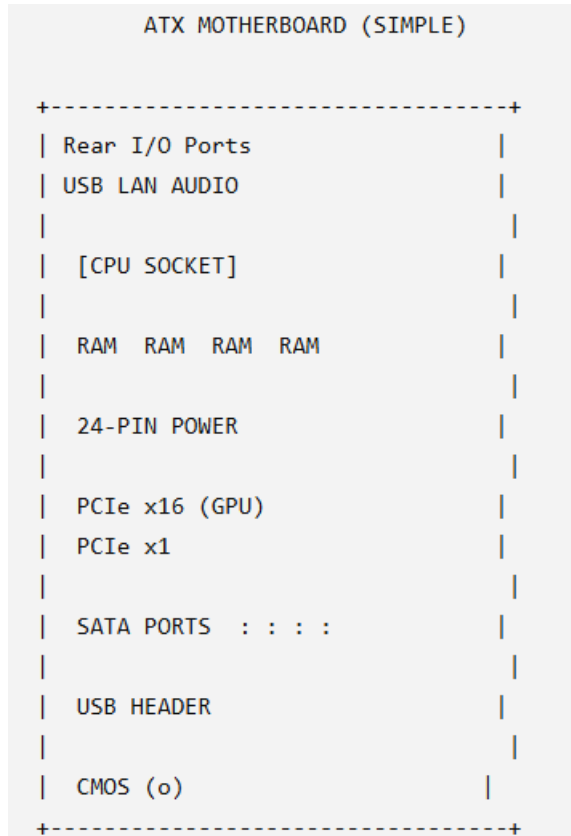


DAY 2

1. Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



- CPU Socket
- RAM Slots
- 24-pin Power Connector
- PCIe x16 Slot
- PCIe x1 Slot
- SATA Ports
- USB Header
- Rear I/O Ports
- CMOS Battery

2. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

LGA 1151 (Intel)

- Compatible CPU: **Intel Core i7-8700K**

- Used in Intel 6th, 7th, and 8th/9th generation processors depending on chipset support.

AM4 (AMD)

- Compatible CPU: **AMD Ryzen 5 3600**
- Widely used socket supporting Ryzen 1000 to 5000 series desktop processors.

LGA 1200 (Intel)

- Compatible CPU: **Intel Core i5-10400**
- Supports 10th and 11th generation Intel Core processors.

3. Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

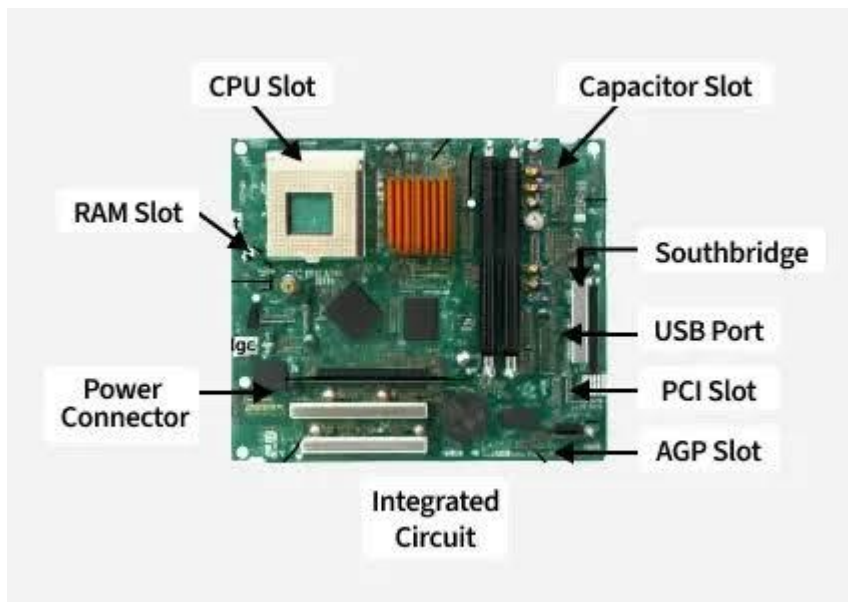
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The CPU acts as the computer's brain, executing software instructions and performing calculations. The motherboard's chipset acts as the central traffic controller, routing data efficiently between the CPU, memory, storage, and peripheral devices

Roles in Action: Opening Spotify

1. **Initiating the Click:** When you click the Spotify icon, your mouse sends an input signal to the chipset, which immediately forwards this event instruction to the CPU
2. **Fetching Data:** The CPU requests the Spotify program files from your SSD or hard drive. This data passes through the chipset (acting as an intermediary) and gets loaded into the fast-access RAM.
3. **Running the App:** The CPU actively processes the software code from RAM to launch the application interface on your screen.
4. **Playing Music:** When you press play, the CPU decodes the compressed audio data from RAM, converts it into an uncompressed audio stream, and sends it back through the chipset. The chipset routes this digital audio signal to your audio hardware (speakers or headphones) so you hear the music.

4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.



CPU Socket (NOT “CPU Slot”)

→ Where the processor is installed

RAM Slots (DIMM slots) (NOT just “RAM Slot”)

→ Where memory sticks are inserted

Power Connector (ATX 20/24-pin)

→ Main power input from SMPS

PCI Slot (Expansion Slot)

→ Used for expansion cards like sound/network cards

USB Ports (Rear I/O / USB header connection)

→ Used to connect USB devices